Niki Nezakati

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Research Interests

- **Diffusion Models for Restoration and Inverse Problems.** Developing diffusion-based generative models for recovering missing or degraded information, with a focus on image restoration, reconstruction, and inverse problems.
- **3D Generation and Rendering.** Developing methods for 3D generation, reconstruction, and rendering, with a focus on visual quality, geometric consistency, and efficiency.
- **Robust and Trustworthy Machine Learning.** Developing machine learning models that remain reliable under missing, noisy, or imperfect data, with a focus on robustness and generalization.
- **Computer Vision and Natural Language Processing.** Exploring vision–language interactions, multimodal representation learning, and cross-domain reasoning at the intersection of CV and NLP.

Skills

(Proficient | Familiar)

Programming Languages : Python, C++, Bash | C/C#, Java, SQL, VHDL, Matlab
Libraries & Frameworks : PyTorch, TensorFlow, Keras | Django-Rest, Django, ASP.NET
Tools & Platforms : Linux, Git, CUDA, OpenCV, COLMAP | Docker, PostgreSQL, SQLite
Compute & Deployment : GPU compute pipelines, cluster/server workflows, Conda, SLURM
Machine Learning & AI : Deep learning, generative AI, computer vision, natural language processing, neural rendering
CS Fundamentals : Data structures, algorithms, software engineering, operating systems
Languages : English (TOEFL iBT 115/120), Persian (Native)

Education

Doctor of Philosophy in Computer Science *2023 – Present*
University of California, Riverside, United States **GPA: 3.91/4**
Co-supervised by Professors Amit K. Roy-Chowdhury and Vishwanath Saragadam

Bachelor of Science in Computer Engineering *2019 – 2023*
Iran University of Science and Technology, Tehran, Iran **GPA: 3.84/4**
Thesis: Generating Answer to Visual Questions in the Medical Domain

Diploma in Mathematics and Physics Discipline *2013 – 2019*
Farzanegan 2 High School, Tehran, Iran **GPA: 4/4**
Operated by the National Organization for Development of Exceptional Talents

Research & Work Experience

Fast High-Fidelity 3D Generation Apple, AI/ML Research Team
Machine Learning Research Intern *Mar. 2026 – Sep. 2026*

- Developed and implemented 2D supervision techniques to improve the quality of few-step 3D generation.
- Trained and optimized flow-based 3D generative models using image-level supervision for faster, higher-quality generation.
- Built evaluation pipelines and benchmarked generation quality and speed against state-of-the-art 3D models.

Robust Diffusion-Based Image Restoration UC Riverside, VLG and CODE Labs
Graduate Research Assistant *Jun. 2025 – Present*

- Developing robust diffusion-based generative models for image restoration under realistic sensor noise.
- Designing robust restoration pipelines for RGB and thermal images using modern diffusion models.
- Improving noise robustness and performance through algorithmic and architectural enhancements.

Robust Multi-Modal Machine Learning UC Riverside, in collaboration with Amazon
Graduate Research Assistant *Sept. 2023 – Sept. 2024*

- Improving robustness of multimodal deep learning models under incomplete or missing modalities.
- Optimizing performance across all input modality combinations via masked-modality training and evaluation.
- Training models on both homogeneous and heterogeneous data types (e.g., vision, language, and structured signals).

Generative Visual Question Answering IUST, Natural Language Processing Lab
Undergraduate Research Assistant *Apr. 2022 – Apr. 2023*

- Working on generative VQA models to generate multi-word answers to visual questions in the medical domain.
- Implemented transformer-based and LSTM-based encoder–decoder architectures with pre-trained CNN encoders.
- Reproduced and extended FSVQA as a baseline, including data preprocessing, training, and evaluation.

Selected Publications

CARD: Correlation Aware Restoration with Diffusion CVPR 2026
Niki Nezakati, Arnab Ghosh, Amit K. Roy-Chowdhury, Vishwanath Saragadam

FLiTo: Fast High-Fidelity 3D Generation via Image Supervision Under Review
Dorian Chan, Niki Nezakati*, Muhammed Kocabas, Vishwanath Saragadam, Oncel Tuzel, Jen-Hao Rick Chang*

Broadband Full Field of View Imaging with Computational Mirrors ECCV 2026
Vishwanath Saragadam, Niki Nezakati, Amit Roy-Chowdhury, Vivek Boominathan

MMP: Towards Robust Multi-Modal Learning with Masked Modality Projection IEEE BigData 2025
Niki Nezakati, Md Kaykobad Reza, Ameya Patil, Mashhour Solh, M. Salman Asif

Honors & Awards

Dean's Distinguished Fellowship Award. University of California, Riverside *2023*

M.Sc. Admission Offer for Outstanding Students. Iran University of Science and Technology *2023*

Undergraduate Tuition Waiver for Exceptional Students. Iran University of Science and Technology *2020*

Ranked Top 0.07% in National University Entrance Exam. Rank 293 of 400,000+ nationwide *2019*

First Place, Junior Demo Open AI & Robotics Challenge. IranOpen International Competitions *2017*

Selected Projects

(Skills | Focus Area)

- **Tile Crack Detection with U-Net.** Python, PyTorch/Keras, OpenCV, NumPy | Image segmentation with convolutional neural networks for detecting cracks in industrial tiles.
- **COVID-19 X-Ray Classification.** Python, PyTorch, Keras, OpenCV | Medical imaging and transfer learning for COVID-19 diagnosis from chest X-rays.
- **Genetic and Ant Colony Algorithms.** Python, NumPy | Optimization with evolutionary algorithms and ant colony optimization on combinatorial problems.
- **Football Match Fuzzy Predictor.** Python, Simpful | Fuzzy logic and rule-based modeling for match outcome prediction.
- **Image Recall with Hopfield Network.** Python, Keras, NumPy, Matplotlib | Associative memory with Hopfield networks for image recall.
- **SOM and RBF Clustering.** Python, TensorFlow/Keras, scikit-learn | Clustering and representation learning using self-organizing maps and radial basis function networks.
- **UC Berkeley CS188 Pac-Man AI.** Python, A*/UCS/DFS/BFS, Q-learning | Search algorithms and multi-agent reinforcement learning.

Teaching Experience

- **University of California, Riverside:** Introduction to Computer Science (Head TA) *(Summer '25)*, Introduction to Machine Learning and Data Mining *(Fall '24)*
- **Iran University of Science and Technology:** Computational Intelligence *(Fall '23)*, Operating Systems *(Spring '22)*, Theory of Languages and Automata *(Spring '22)*, Algorithms Design *(Spring '22)*, Data Structures *(Fall '21)*, Advanced Programming *(Spring '21)*, Discrete Mathematics *(Spring '21)*

Related Coursework

- **Artificial Intelligence and Machine Learning:** Deep Learning (A+), Natural Language Processing (A+), Computer Vision (A+), Computational Intelligence (A+).
- **Data Science and CS Fundamentals:** Data Structures (A+), Databases (A+), Cybersecurity (A+), Signal Processing (A+), Operating Systems (A+), Data Communication (A+), Advanced Programming (A+), Differential Equations (A+).

Online Courses

- **Natural Language Processing with Deep Learning** – Christopher D. Manning, Stanford University.
- **Improving Deep Neural Networks** – Andrew Ng, DeepLearning.AI.
- **Neural Networks and Deep Learning** – Andrew Ng, DeepLearning.AI.
- **Advanced Algorithms and Complexity** – Alexander Kulikov et al., UC, San Diego.
- **Algorithms on Strings** – Alexander Kulikov et al., UC, San Diego.
- **Algorithms on Graphs** – Alexander Kulikov et al., UC, San Diego.
- **Data Structures** – Alexander Kulikov et al., UC, San Diego.
- **Algorithmic Toolbox** – Alexander Kulikov et al., UC, San Diego.
- **Python Data Structures** – Charles Russell Severance, University of Michigan.
- **Using Python to Access Web Data** – Charles Russell Severance, University of Michigan.
- **Web Application Technologies and Django** – Charles Russell Severance, University of Michigan.

References

Amit K. Roy-Chowdhury **Professor, Presidential Chair | amitrc@ucr.edu** **Website**

Vishwanath Saragadam **Assistant Professor | vishwans@ucr.edu** **Website**